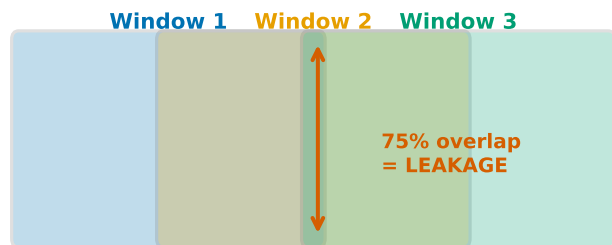


When Benchmarks Lie: Temporal Data Leakage Reverses Time Series Anomaly Detection Rankings

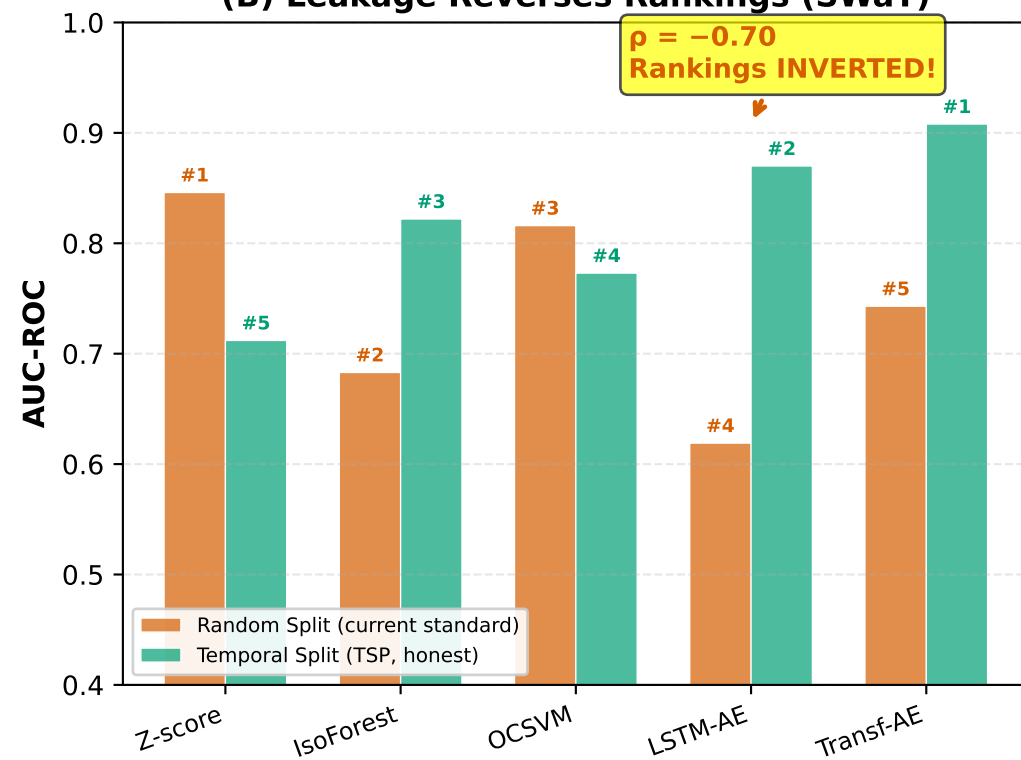
(A) Random Split on Sliding Windows

Raw Time Series

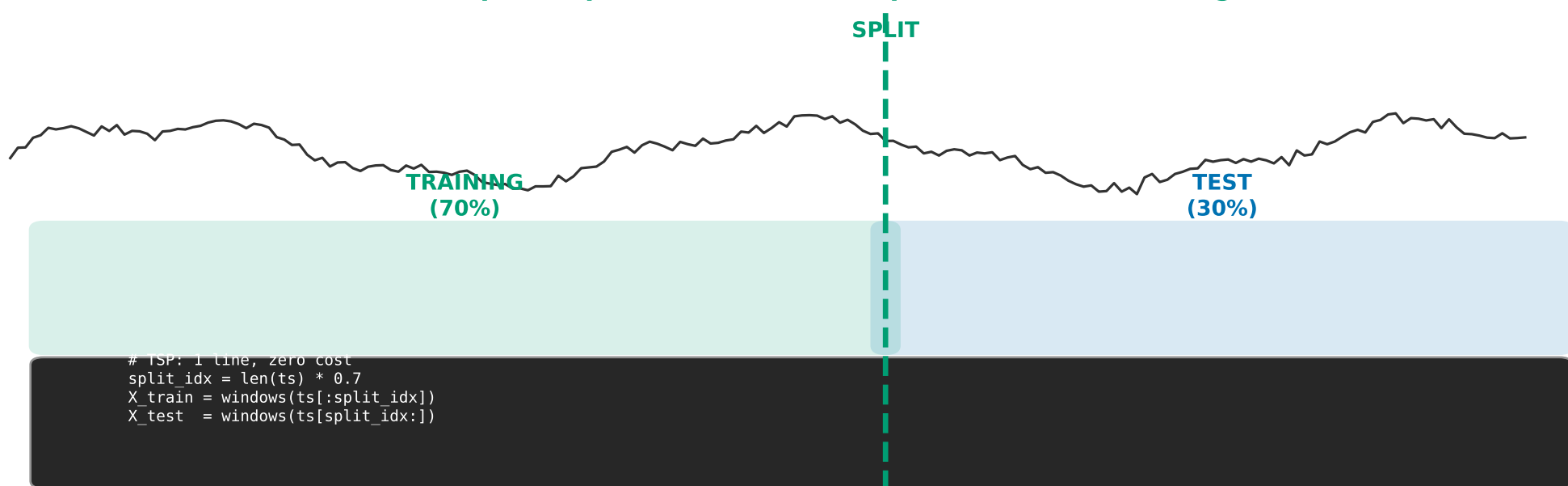


Random Split → **Window 2 in TRAIN, Window 3 in TEST**
→ Model has already seen 75% of test data!

(B) Leakage Reverses Rankings (SWaT)



(C) Temporal-Split Protocol (TSP): Split BEFORE Windowing



Proposed fix: Temporal-Split Protocol (TSP) — one line of code, zero new infrastructure, eliminates the largest source of TSAD evaluation inflation